Project Plan

Project Title:
Master for an Hour – Call Control System for Minor Repairs

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1. Users Definition:

The application will serve the following types of users:

- **Customer:**
  - **Responsibilities:**
    - Create repair service requests.
    - Select the type of repair (e.g., plumbing, electrical work, painting, furniture assembly, etc.).
    - Provide details such as address and preferred service date/time.
    - Track the status of their repair requests.
    - View history of completed repairs.
- **Operator (Dispatcher):**
  - **Responsibilities:**
    - Review incoming repair requests.
    - Assign masters to specific tasks based on availability and skill.
    - Update request statuses (e.g., New, Assigned, In Progress, Completed, Cancelled).
    - Manage the list of available masters, including their status and performance.
- **Master (Worker):**
  - **Responsibilities:**
    - View and manage assigned repair tasks.
    - Update task progress (mark as in-progress, completed, etc.).
    - Add notes related to completed tasks (e.g., issues encountered, recommendations).
    - Ensure communication with both customers and operators as needed.

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2. Functionality Definition:

The application will support the following core features:

- **User Registration and Authentication:**
  - Allow customers, operators, and masters to register with the system and log in securely.
- **Repair Request Management:**

    - Enable customers to create new repair requests by specifying type, description, address, and preferred date/time.
    - Allow customers to modify or cancel requests before assignment.
- **Request Assignment and Status Updates:**
    - Operators can assign repair tasks to available masters based on expertise and availability.
    - Operators and masters can update the status of requests (e.g., New, Assigned, In Progress, Completed).
    - Customers will be notified of any changes in request status.
- **Repair History:**
    - Customers and operators can view a history of completed repairs, including details like service type, date, and task completion status.
- **Data Storage and Management:**
    - Store repair requests and task data in a **Supabase** database for persistence and scalability.
    - Provide an option to save and load data for requests, including users and task progress.
- **Input Validation and Error Handling:**
    - Ensure all user inputs (address, date/time, repair description) are validated and error-free.
    - Display clear error messages to guide users in case of incorrect or missing information.
- **Basic Statistics and Reports:**
    - Provide operators with statistics such as the number of completed jobs, active requests, and task performance metrics.

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3. Platform:

- **Web Application** designed for seamless access from desktop and mobile devices in the future.
- **Frontend:** Django (with Python and JavaScript)
- **Backend:** Django (Python) with **Supabase** for real-time database handling.

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4. Programming Languages and Tools:

- **Primary Language:** Python
- **Framework:** Django for backend management
- **Database:** Supabase (for data storage and user authentication)
- **Frontend:** Django templates for user interface
- **Development Paradigm:** Object-Oriented Programming (OOP)